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Robust WiFi Sensing-based Human Pose Estimation Using Denoising Autoencoder and CNN with Dynamic Subcarrier Attention

Xuan Hoang Nguyen, Van-Dinh Nguyen, *Senior Member, IEEE*, Quang-Trung Luu, Toan Dinh Gian, and Oh-Soon Shin, *Senior Member, IEEE*

***Abstract*—WiFi sensing-based human pose estimation (HPE) has gained significant attention in the academic community due to its advantages over vision- and sensor-based methods, including non-intrusiveness, convenience, and enhanced privacy protection. However, most existing WiFi-based pose estimators suffer from poor performance and lack robustness in the presence of random noise. To address these challenges, this paper presents a novel HPE architecture comprising two key modules: A Denoiser and an Estimator. The Denoiser is based on an autoencoder structure, while the Estimator is based on a new convolutional neural network (CNN) called SDy-CNN, which is designed to dynamically focus on high-information subcarriers of orthogonal frequency division multiplexing signals. Additionally, Bayesian optimization is employed to fine-tune the architecture's parameters for optimal performance flexibly. Experiments conducted on a comprehensive dataset, MM-Fi, demonstrate that the proposed architecture significantly outperforms existing state-of-the-art estimators, achieving up to an 8.38% improvement in HPE accuracy in clean data scenarios and up to a 14% improvement in noisy data scenarios. It has also been proven to gain computational efficiency when being much faster than other methods.**

***Index Terms*—Bayesian optimization, dynamic convolution, denoising autoencoder, human pose estimation, transfer learning, WiFi sensing.**

I. Introduction

A. Motivation

HUMAN pose estimation (HPE) involves detecting and estimating the positions of key-points on the human body, such as joints (*e.g.*, elbows, knees, shoulders) [1], [2]. This technology has a wide range of practical applications, including virtual reality [3], human-robot interaction [4], healthcare [5], and power station inspections [6]. However, most existing HPE approaches primarily rely on cameras [7], [8] or wearable devices [9], [10], each with inherent limitations. Camera-based HPE systems often struggle with privacy concerns, lighting conditions, and blind spots, while wearable devices can be cumbersome and inconvenient. Additionally, both methods tend to be relatively expensive. As a result, there is a growing interest in exploring more convenient, non-invasive, and cost-effective alternatives, such as WiFi sensing-based HPE.

Recently, WiFi sensing technology has emerged as a promising solution for HPE thanks to its cost efficiency, privacy preservation, and device-free wireless sensing capabilities [6], [11]. By leveraging channel state information (CSI) from ubiquitous WiFi devices in indoor environments, especially in poorly lit rooms, WiFi sensing-based estimators can effectively predict human poses. CSI captures variations in wave propagation across individual subcarriers in the WiFi channel based on orthogonal frequency vision multiplexing (OFDM), providing useful yet implicit information about the relationship between WiFi signal propagation and human body poses within the environment. Despite its advantages, WiFi sensing still faces significant challenges, including limited spatial resolution of CSI data, variability in environmental conditions (*e.g.,* walls, furniture, and other obstructions), and hardware dependency (*e.g.,* the number, placement, and configuration of WiFi transmitters and receivers). These factors can greatly affect sensing accuracy and reliability, highlighting the need for further research into advanced methods to enhance the effectiveness and robustness of WiFi-based human pose estimation systems.

To extract meaningful spatial features from CSI data for predicting human key-points, deep neural networks, such as convolutional neural networks (CNNs), have been taken into consideration widely in the literature [12]–[16]. These existing CNN models typically learn spatial patterns from CSI data using two-dimensional kernels with fixed parameters after the training process. However, CSI patterns are often implicit [17], and the relevance of features in each data dimension may vary depending on their spatial positions. Consequently, CNN-based extractors may cause the learning model to overlook information-rich regions, leading to suboptimal feature extraction and, ultimately reduced performance of HPE models. Recent studies, such as [11], [18], have attempted to address this issue by employing the multi-head self-attention mechanism, originally introduced in the Transformer model [19]. While this mechanism is designed to extract more fine-grained information by focusing on information-dense subcarriers, it is more suited for long sequential data rather than CSI data. This is attributed to the fact that each data sample for WiFi sensing-based HPE corresponds to a CSI packet with a relatively short

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duration. Additionally, the self-attention mechanism often demands substantial computational resources, making it challenging to deploy on resource-constrained devices. Therefore, there is a pressing need for a more efficient and lightweight attention mechanism tailored for WiFi sensing-based HPE to achieve high performance.

Real-world CSI patterns, on the other hand, are highly susceptible to excessive noise [20]. Despite the utilization of denoising techniques, such as traditional signal processing filters, a high noise-to-feature ratio can persist due to the subtle nature of human pose-related signals and the presence of unpredictable interference (random noise) [16], [17]. Moreover, most human pose estimators in the literature have been trained on cleaner data that have typically undergone extensive pre-processing. As a result, their performance in HPE may deteriorate when deployed in real-world environments with high levels of random noise. This calls for a more effective and robust denoising method.

B. Main Contributions

In this work, we propose a robust HPE architecture to address the aforementioned challenges. The proposed architecture takes the CSI of WiFi signals as input and consists of two primary modules: i) An autoencoder (AE)-based **Denoiser** and ii) an **Estimator** for key-point prediction. Specifically, the Denoiser employs a CNN to reduce the dimensionality of noisy CSI data while preserving essential features, making it effective for both noise removal and capturing relevant patterns for HPE. Dimensionality reduction is crucial because higher-dimensional data can significantly increase computational complexity [21]. For the Estimator, we introduce a novel attention network called "SDy-CNN," specifically designed to extract features more effectively from CSI data. In this context, certain subcarriers that carry little pose-related information can degrade the network's performance while increasing computational costs [22]. To address this, SDy-CNN selectively focuses on subcarriers that are most relevant to human poses, thus reducing computational resource consumption and enhancing HPE performance. In addition, such dynamic feature adaptation enables the Estimator to recalibrate in real time, effectively handling variations in environmental conditions and user behavior without requiring very frequent retraining.

Hyperparameter optimization is crucial for maximizing model performance, efficiency, and adaptability, ultimately leading to more accurate and reliable predictions. However, this aspect is often overlooked in studies on WiFi sensing-based HPE. Most of the previous studies rely on experience or heuristic methods to set model hyperparameters, which seldom ensures optimal performance. To address this limitation, we employ the Bayesian optimization (BO) algorithm [23] to identify the optimal hyperparameters for the proposed architecture, thereby enhancing both performance and configuration flexibility. To sum up, our main contributions are as follows:

- We propose a new and robust HPE architecture that introduces a Denoiser based on an AE to address the challenges of random noise and high dimensionality in raw CSI data. This module not only effectively filters out random noise but also reduces data dimensionality, ensuring that the most relevant features are retained for subsequent processing.
- We develop an Estimator that is a lightweight yet dynamic CNN network. This network is designed to intelligently prioritize and extract features from CSI subcarriers that contain the most critical and relevant information for HPE tasks. By focusing on these key subcarriers, the proposed Estimator enhances accuracy and efficiency while minimizing computational overhead and retraining frequency.
- We employ the BO algorithm to dynamically search for the optimal hyperparameters for our proposed HPE architecture. This approach allows for efficient exploration of the hyperparameter space and versatile identification of configurations that maximize the model performance.
- Numerical results show that the proposed HPE approach significantly outperforms existing state-of-the-art (SOTA) models across several key performance metrics. Specifically, our estimator achieves significantly higher accuracy, greater resilience to random noise, and improved time efficiency, underscoring its superior performance in HPE compared to existing state-of-the-art approaches.

The remainder of this paper is organized as follows. In Section II, we provide an overview of relevant works in HPE. The background knowledge is then presented in Section III. Section IV elaborates on the proposed robust WiFi sensing-based HPE architecture, followed by performance evaluation and experiments in Section V. Finally, conclusions and directions for future work are discussed in Section VI.

II. Related Work

WiFi sensing has emerged as a promising alternative to traditional sensing technologies, such as cameras and wearable devices, due to its non-intrusive nature, cost efficiency, and independence from physical equipment. Extensive research has explored various WiFi sensing applications, including human identification [24], [25] and gesture recognition [26]. While studies [27], [28] have demonstrated the potential of deep learning models to enhance human localization using WiFi signals, [17] is recognized as a leading approach for WiFi-based human activity recognition. Along with these advancements, leveraging commodity WiFi for effective and convenient HPE remains a highly promising area compared to camera-based or wearable device methods.

WiFi sensing-based HPE: Significant progress has been made in the field of HPE using WiFi sensing. The authors in [12] explored the potential of WiFi for identifying single-person poses by employing a fully convolutional network called WiSPPN to extract features from CSI data. Achieving an accuracy exceeding 75%, WiSPPN was tested on a private dataset collected from eight volunteers in laboratory and classroom environments. Building on a similar approach, the work [29] utilized a CNN to reconstruct 2D spatial information of human bodies. However, both studies were limited by their use of data from only 30 subcarriers, resulting in low-resolution data and then suboptimal performance. Furthermore, such conventional CNNs with fixed-size kernels, as also seen in [15],

TABLE I: Summary of Recent Studies on WiFi Sensing-based HPE Tasks.

Refs.	Learning Model	Type of Dataset	Hyperparameter Selection	Versatile Configuration	Raw Data	Year
[12]	FCN	Private	Heuristic	×	×	2019
[29]	U-Net	Private	Heuristic	×	×	2019
[14]	CNN + LSTM	Private	Heuristic	×	×	2020
[16]	CNN	Private	Heuristic	×	×	2020
[30]	CNN	Private	Heuristic	×	×	2021
[15]	CNN	Private	Heuristic	×	×	2021
[31]	DINN	Private	Heuristic	×	×	2021
[22]	Performer + CNN	Public	Heuristic	×	✓	2022
[18]	Performer + CNN	Public	Heuristic	×	✓	2022
[11]	CNN + Transformer	Public	Heuristic	×	×	2023
[6]	Conformer	Private	Heuristic	×	×	2024
Ours	Autoencoder + SDy-CNN	Public	Bayesian Optimization	✓	✓	—

[16], [30], [31], struggle with capturing spatial generalization and fine-grained CSI features. This limitation can impact the accuracy of the learning models, reducing their practicality in real-world applications. To address these challenges, Jiang *et al.* [14] incorporated long short-term memory (LSTM) with CNNs to enhance the capture of deep CSI information. Despite this advancement, the approach still relies heavily on the traditional CNN structure for feature extraction, while LSTM is resource-intensive and may not fully mitigate the shortcomings of conventional CNN approaches.

Self-attention mechanism: To further improve performance across various applications, recent studies on WiFi sensing-based HPE have adopted the self-attention mechanism introduced in [19]. For example, the work in [11] developed MetaFi++, which uses Transformers on top of three CNNs with shared weights to extract richer spatial information from standard WiFi devices, enabling virtual character simulation in the Metaverse. Similarly, the authors in [6] introduced the PowerSkel framework, which applies the self-attention mechanism to extract features from CSI data in power station environments. This approach showed high performance on a private dataset, highlighting the potential of WiFi sensing for enhancing personnel safety in electricity scenarios. However, the self-attention mechanism is generally more suitable for long sequential data [32], making it less compatible with CSI data used in HPE, in which each sample typically represents a relatively short duration. Additionally, the self-attention mechanism often incurs significant computational overhead, which could limit the practicality of these human pose estimators in real-world applications.

In addition to feature extraction, some studies have also employed the self-attention mechanism to mitigate environmental interference [18], [22]. These approaches aim to design estimators that focus on subcarriers that are less prone to noise. While these methods have achieved commendable accuracy on their own Wi-Pose dataset, their denoising effectiveness requires further validation on more reliable, diverse, and comprehensive datasets. Moreover, the limitations associated with multiple multi-head self-attention layers in these studies, such as increased computational complexity and potential incompatibility with shorter data sequences, need careful consideration.

Our work: To address the aforementioned shortcomings, we propose a robust and dynamic architecture designed to withstand random noise in raw data during practical deployments while still effectively extracting CSI features using dynamic kernels. Unlike traditional CNNs with fixed kernels, our SDy-CNN dynamically adjusts its kernels to the input characteristics, enabling adaptation to high-information subcarriers and capturing subtle patterns critical for human pose estimation. Additionally, compared to computationally expensive self-attention mechanisms, SDy-CNN employs a lightweight, attention-like mechanism tailored for short-duration CSI signals. This approach significantly reduces computational overhead while prioritizing information-dense subcarriers, efficiently prioritizing the design for resource-constrained applications. We also incorporate the BO algorithm to strike optimal performance. Unlike heuristic methods, BO in our design systematically identifies optimal hyperparameters and allows flexible adjustment of the model's depth. In summary, Table I provides an overview of WiFi sensing-based HPE studies in the literature, comparing our architecture with existing approaches across various criteria.

III. Background

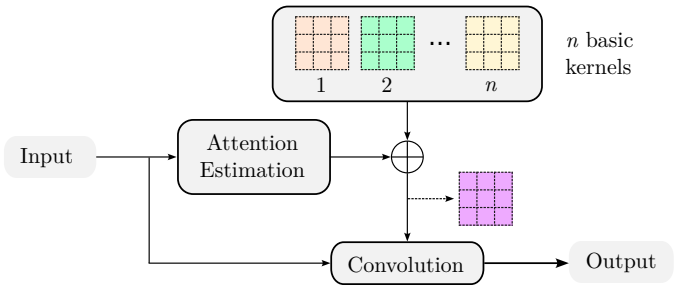


Fig. 1: The general procedure of dynamic convolution.

A. Channel State Information

In wireless communication, CSI represents the channel characteristics of a communication link. Given $\boldsymbol{x}$ as the WiFi signal transmitted from a transmitter and $\boldsymbol{y}$ as the signal obtained by a receiver, the received signal can be represented as

$$\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x} + \boldsymbol{\eta} \tag{1}$$

where $\boldsymbol{H}$ is the complex channel matrix containing CSI data for the subcarriers in an OFDM system and $\boldsymbol{\eta}$ is the channel

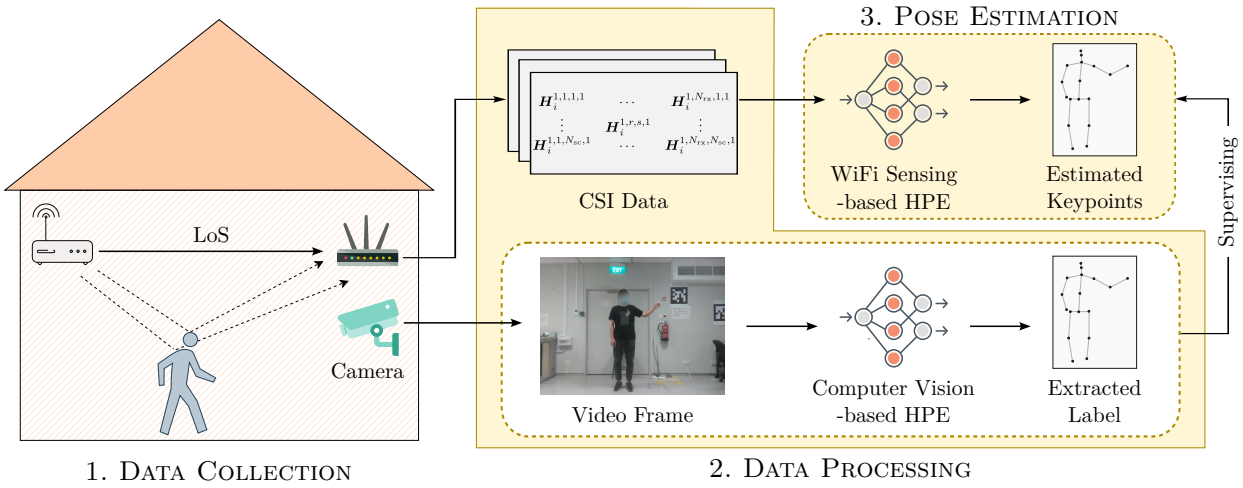


Fig. 2: The WiFi sensing-based HPE pipeline.

noise. The CSI value $\boldsymbol{H}_i(f,t)$ for each subcarrier i with frequency f at time t is given by [33]

$$\boldsymbol{H}_i(f,t) = e^{-\jmath 2\pi \Delta f t}\Big(\boldsymbol{H}_s(f) + \sum_{p=1}^{P} \alpha_p(f,t) e^{\frac{-\jmath 2\pi d_p(t)}{\lambda}}\Big) \quad (2)$$

where

- $e^{-\jmath 2\pi \Delta f t}$ represents the phase shift due to carrier frequency offset, packet detection delay, and sampling frequency offset;
- $\boldsymbol{H}_s(f)$ is the CSI reflected from static objects or transmitted along the line-of-sight path;
- $\alpha_p(f,t)$ is the complex attenuation factor and initial phase offset of the p-th transmission path with N being the number of paths;
- $e^{\frac{-\jmath 2\pi d_p(t)}{\lambda}}$ is the phase shift on the p-th path, with $d_n(t)$ being its length and λ being the signal wavelength.

In real-world channel environments, phenomena such as wave reflection, diffraction, and scattering frequently occur, causing fine-grained variations in CSI data. These variations can be harnessed to reveal the underlying relationship between signal patterns and HPE [20].

B. Autoencoder

A conventional autoencoder is an unsupervised neural network designed to reconstruct its output to match the input data closely. It generally consists of three components: encoder, code, and decoder. When the input data is fed to the AE network, the encoder part compresses it into a lower-dimensional code. This code retains the most crucial information of the input data while having fewer features. Meanwhile, with a similar structure to the encoder, the decoder part is in charge of reconstructing the original input from the code. Given the input and output of the AE model, denoted as $\boldsymbol{X}$ and $\boldsymbol{X}'$, respectively, the mapping function from $\boldsymbol{X}$ to $\boldsymbol{X}'$ can be described as $\boldsymbol{X}' = \mathit{Decoder}\big(\mathit{Encoder}(\boldsymbol{X})\big)$. The model's weights are adjusted to minimize the reconstruction error $\mathcal{L}$, defined as follows:

$$\mathcal{L}(\boldsymbol{X}, \boldsymbol{X}') = \left\|\boldsymbol{X} - \boldsymbol{X}'\right\|^2. \quad (3)$$

By leveraging the inherent properties of a standard AE, we can achieve two key objectives: Dimensionality reduction and denoising. It is important to note that the primary purpose of an AE is to mitigate overfitting and ensure the model learns meaningful representations, it also effectively aids in denoising.

C. Dynamic Convolution

Deep neural networks have been widely adopted across various tasks, primarily due to the effectiveness of convolutional layers. The core component of a typical convolutional layer is the *kernel*. Once optimized during the training process, the kernel values remain fixed, allowing each kernel to extract only a specific feature. As a result, numerous kernels are required to capture a diverse range of features, which, while enhancing model performance, also introduces computational burdens and optimization challenges.

To address these issues, a kernel-combining approach called dynamic convolution was introduced by [34]. As shown in Fig. 1, a dynamic convolution generates a new content-adaptive kernel by assembling several basic kernels in parallel, guided by attention weights derived from the input. This input-aware mechanism allows dynamic convolution to adjust multiple kernel values even after the training process, making it computationally efficient and more versatile in feature representation than conventional CNNs.

Owing to these advantages, dynamic convolution has been adapted to various applications, including text-independent speaker recognition [35], human recognition [36], and sound event detection [37]. Inspired by the structure of dynamic convolution, we have designed a dynamic network tailored for WiFi sensing-based HPE, effectively and flexibly extracting CSI features while overcoming the limitations of traditional CNNs and the self-attention mechanism.

IV. Proposed Architecture

In this section, we first introduce briefly the WiFi sensing-based HPE architecture, and then detail the design of the Denoiser and Estimator, followed by the hyperparameter optimization process.

A. Overview of the HPE Architecture

A WiFi sensing-based HPE system typically comprises three main phases: data collection, data processing, and pose estimation. As illustrated in Fig. 2, both a camera and a pair of WiFi devices are deployed concurrently during the data collection phase to capture video frames and CSI data, respectively. These two types of data are subsequently processed for HPE. In the data processing phase, the video frames are fed into a computer vision-based HPE model to extract key-point coordinates, which serve as labels to guide the training of a WiFi sensing-based HPE model. Concurrently, the raw CSI data undergoes a cleaning process to remove phase values, which are highly susceptible to noise. The cleaned amplitude values are then passed to the final phase, *i.e.*, pose estimation. During this phase, the CSI data is further refined, and its features are extracted by an HPE, leading to the prediction of key-point coordinates as the final output.

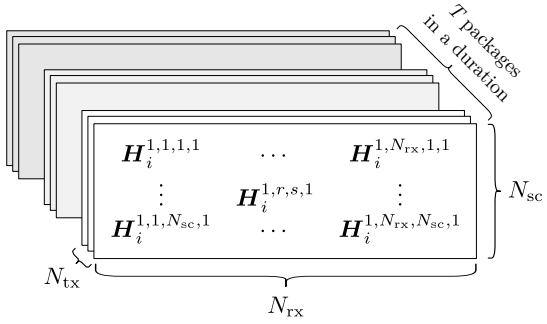


Fig. 3: The dimensional diagram of a CSI frame.

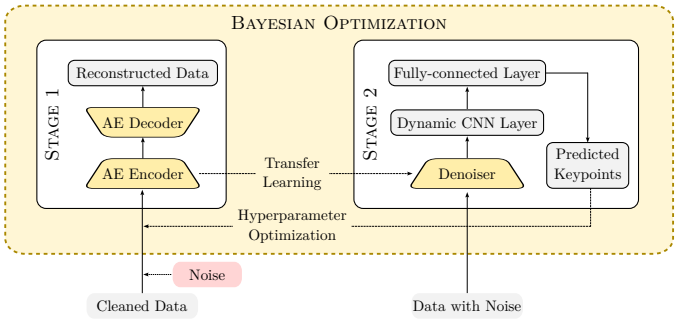


Fig. 4: The proposed WiFi sensing-based HPE architecture.

This study focuses on designing an effective and dynamic architecture for the final phase. Let us consider the CSI dataset of N samples $\{\boldsymbol{H}_i\}_{i\in[1,N]}$ where $\boldsymbol{H}_i \in \mathbb{R}^{N_{\text{tx}}\times N_{\text{rx}}\times N_{\text{sc}}\times T}$. As illustrated in Fig. 3, the dimensions denote the numbers of transmit antennas N_{tx}, receive antennas N_{rx}, subcarriers N_{sc}, and CSI packets T within a specific duration. The output of each sample $\boldsymbol{H}_i$ is the localized key-point values $\widehat{\boldsymbol{y}}_i$. Our objective is to accurately estimate $\hat{Y}_i$ from $\boldsymbol{H}_i$:

$$\widehat{\boldsymbol{y}}_i = f\left(\boldsymbol{H}_i\right). \tag{4}$$

To achieve this, we propose a novel architecture for WiFi sensing-based HPE, as illustrated in Fig. 4.

The architecture is composed of two main stages. The first one involves training a Denoiser for the denoising task, followed by training a human pose Estimator specifically tailored for the WiFi sensing-based HPE task. More specifically, in the first stage, with the amount of cleaned data from the data processing phase, we aim to develop a robust Denoiser using an AE. This network is trained by taking the noise-corrupted data as input, producing the output with as little noise as possible. After the training process, only the AE's encoder part is retained as the Denoiser through transfer learning. In the second stage, an Estimator, consisting of a dynamic CNN and a fully connected (FC) layer, is trained by taking the Denoiser's output as input. It is noted that the Denoiser's pre-trained weights are kept frozen in this stage. This allows the Estimator to learn informative representations from the denoised CSI data in a lower-dimensional space. In this way, when being deployed, the proposed architecture can effectively handle the input with random noise while still estimating key-point values accurately. In addition, the BO algorithm is applied to both the AE and the Estimator for optimal hyperparameter tuning. In the next sections, the designs of the Denoiser and the Estimator will be presented in more detail.

B. Denoiser

Raw CSI data collected from indoor environments are highly susceptible to excessive random noise [20]. Despite advancements in denoising technologies using traditional filters, eliminating noise remains a significant challenge. This limitation hampers the Estimator's ability to capture essential pose-related features, thereby reducing its effectiveness and feasibility in practical applications. To address this, it is crucial to remove noise from CSI data radically before inputting it into the Estimator. Additionally, in real-world environments, CSI data are often high-dimensional, containing complex and implicit information [38]. Consequently, reducing the dimensionality of the data and eliminating irrelevant or insignificant features are critical. This not only reduces the complexity of the WiFi sensing-based HPE architecture but also enhances its overall performance.

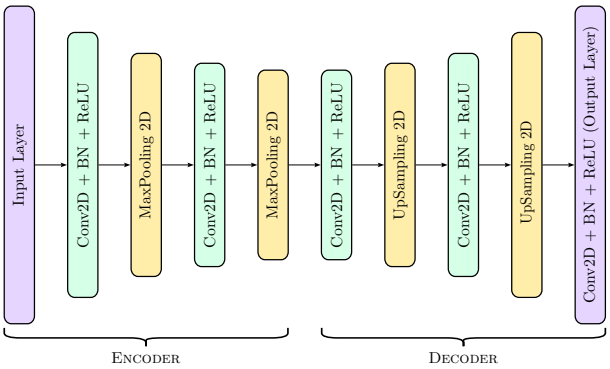


Fig. 5: The structure of AE-based of the Denoiser (Stage 1).

To address both noise reduction and dimensionality reduction, we now develop the Denoiser based on an AE network. As shown in Fig. 5, the network scheme comprises an encoder and a decoder. For computational efficiency and simplicity, we utilize only two convolutional hidden layers in each component. In the encoder, each convolutional layer is followed by a max-pooling layer, while the decoder replaces

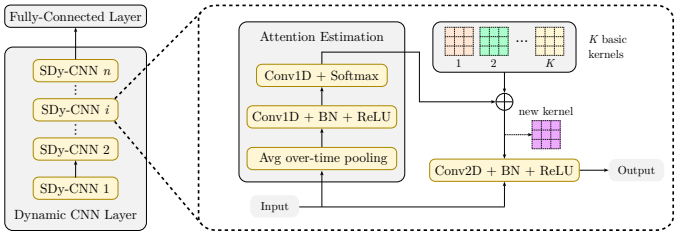


Fig. 6: The structure of the Estimator (Stage 2).

these with upsampling layers. The network concludes with a convolutional output layer to reconstruct the input data. After the training process, only the encoder is retained to function as the Denoiser. Additionally, square kernels are applied to all the convolutional and max-pooling layers in the network. To be more specific, the filter sizes of the two convolutional layers in the encoder are denoted by *ae_kernel_size_1* and *ae_kernel_size_2*, with the corresponding numbers of filters being *n_ae_kernels_1* and *n_ae_kernels_2*, respectively. These hyperparameters are symmetrically configured in the decoder. The two max-pooling layers in the encoder have sizes *ae_max_pool_1* and *ae_max_pool_2*, respectively. Since all of these hyperparameters can influence denoising performance, it is required to set them up thoroughly.

Let the CSI data collected from N_{tx} transmit antennas and N_{rx} receive antennas with N_{sc} subcarriers in a frame of length T be denoted as $\boldsymbol{H}_i(t) = \{\boldsymbol{H}_i^1, \boldsymbol{H}_i^2, \cdots, \boldsymbol{H}_i^T\} \in \mathbb{R}^{N_{\text{tx}} \times N_{\text{rx}} \times N_{\text{sc}} \times T}$. To be computationally compatible with the two-dimensional convolution, the CSI's dimension should be converted to $N_{\text{c}} \times N_{\text{sc}} \times T$, in which $N_{\text{c}} = N_{\text{tx}} \times N_{\text{rx}}$ denotes the number of channels. The corresponding output of the AE network is $\boldsymbol{H}_i' \in \mathbb{R}^{N_{\text{c}} \times N_{\text{sc}} \times T}$. To optimize $\boldsymbol{H}_i'$ to be as close as possible to $\boldsymbol{H}_i$, we utilize the loss function $\mathcal{L}_{\text{denoiser}}$ as defined in (3). Additionally, we adopt a learning rate adjustment strategy to enhance the training process:

$$\zeta_{\text{denoiser}} = \begin{cases} \zeta_0, & \text{if \#epochs} \leq 45, \\ \zeta_0 e^{-0.1}, & \text{otherwise} \end{cases} \tag{5}$$

where ζ_{denoiser} denotes the learning rate function, and ζ_0 is the initial learning rate. After training the AE, the CSI data is fed through the encoder part (*i.e.*, the Denoiser), producing cleaned and compressed codes $\boldsymbol{C}_i \in \mathbb{R}^{N_{\text{c}}' \times N_{\text{sc}}' \times T'}$. These codes are used for training the Estimator in the second stage.

C. Estimator

The Estimator consists of a dynamic layer and FC layer. The former is responsible for CSI feature extraction, while the latter predicts key-point coordinates. As illustrated in Fig. 6, the dynamic CNN layer is constructed by sequentially stacking multiple networks called SDy-CNN, which stands for subcarriers-focusing dynamic convolutional neural network. This novel attention network is designed to extract CSI features more effectively than traditional CNNs or self-attention-based networks. In CSI data, certain subcarriers carry minimal pose information and contribute little to the network's performance while significantly increasing computational costs [22]. To address this, SDy-CNN focuses on subcarriers that are more sensitive to human poses, prioritizing those rich in information. This network is based on the structure of dynamic convolution [34], which has been shown to enhance representation capabilities with manageable computational demands. Furthermore, CSI data heavily depend on environmental conditions [39], leading to variations over time. SDy-CNN's input-aware adaptability enables real-time feature extraction that adjusts to these changes, including user movements. Unlike traditional CNNs with fixed kernels after training, SDy-CNN recalibrates its feature extraction for each input, thus minimizing the need for frequent model retraining.

For convenience, we denote the number of SDy-CNN networks as *n_stacks*. This hyperparameter can significantly impact the Estimator's ability to capture critical CSI features from the data. To achieve an acceptable trade-off between model complexity and performance, *n_stacks* is dynamically determined using the BO algorithm. The dynamic CNN layer operates as follows. Firstly, the initial SDy-CNN network takes the Denoiser's output $\boldsymbol{C}_i \in \mathbb{R}^{N_{\text{c}}' \times N_{\text{sc}}' \times T'}$ as input and computes attention weights using a strategy similar to the squeeze-and-excitation strategy [40]. Specifically, an average-over-time pooling is applied to the input data, transforming the code $\boldsymbol{C}_i$ into the captured information $\boldsymbol{Z}_{i,1} \in \mathbb{R}^{N_{\text{c}}' \times N_{\text{sc}}'}$, which is calculated as

$$\boldsymbol{Z}_{i,1} = \text{Avg}(\boldsymbol{C}_i) = \frac{1}{T'} \sum_{t=1}^{T'} \boldsymbol{C}_i(t). \tag{6}$$

This squeeze part emphasizes the frequency components of the CSI subcarriers. The subsequent excitation stage employs two 1D convolution sublayers rather than fully connected ones as in previous works [34], [40]. This choice is motivated by the effectiveness of 1D convolutions in learning domain-specific features [37], especially for those derived from adjacent CSI frequency values. Furthermore, following the widely-used CNN design practices, we also adopt a batch normalization (BN) operation and one activation function right after the 1D convolution. The computational procedure can be expressed as

$$\beta_{i,1} = \text{ReLU}\big(\text{BatchNorm}(\text{Conv1D}(\boldsymbol{W}_{1,1}, \boldsymbol{Z}_{i,1}))\big) \tag{7}$$

where $\beta_{i,1}$ is the output of the first 1D convolution sublayer in the first SDy-CNN, and $\boldsymbol{W}_{1,1}$ is its weight matrix. The second 1D convolution sublayer assigns an attention scalar to each subcarrier, representing its relative importance. The attention scalar in the first SDy-CNN is computed as

$$\alpha_{i,1} = \text{Softmax}(\text{Conv1D}(\boldsymbol{W}_{2,1}, \beta_{i,1}), \tag{8}$$

where $\boldsymbol{W}_{2,1}$ is its weight matrix.

The attention scalar is then used to dynamically generate a new kernel $\kappa_{i,1}$ by aggregating K basic kernels ($K \in \mathbb{Z}$, $K \geq 2$), defined as

$$\kappa_{i,1} = \sum_{k=1}^{K} \alpha_{i,1}.\boldsymbol{W}_{k,1} \tag{9}$$

where $\boldsymbol{W}_{k,1}$ is the weight matrix of the k-th basic kernel. This allows the SDy-CNN to prioritize informative subcarriers using data-driven attention weights, while simultaneously

adapting to variations in input data in real time. The feature map $\boldsymbol{C}'_{i,1}$ from the first SDy-CNN network is given as

$$\boldsymbol{C}'_{i,1} = \text{ReLU}\big(\text{BatchNorm}(\text{Conv2D}(\kappa_{i,1}, \boldsymbol{C}_i))\big). \tag{10}$$

If *n_stacks* is 2 or greater, $C'_{i,1}$ is fed into the next SDy-CNN, and the process is repeated as described above. Finally, the last CSI feature map $\boldsymbol{C}'_{i,n_stacks} \in \mathbb{R}^{N''_{\text{c}} \times N''_{\text{sc}} \times T''}$ is fed into the FC layer to predict human pose coordinates, as illustrated in Fig. 4. The Estimator is trained using the mean squared error (MSE) as the loss function.

D. Hyperparameter Optimization

Since CSI data is highly sensitive to environmental conditions, an effective HPE model must adapt to shifts in data patterns. Regularly re-training the model by updating its hyperparameters is therefore essential, as these parameters significantly affect its performance. However, previous works, such as [12], [16], [29], [31], often rely on heuristic methods or expert intuition to determine hyperparameter values. These approaches are not only time-consuming and labor-intensive but also unsuitable for handling the dynamic nature of CSI data. Instead, an HPE should be optimized automatically to accommodate changing CSI patterns over time, rather than relying on manual adjustments.

To ensure the proposed architecture functions dynamically and effectively in diverse practical settings, we utilize the BO algorithm to find optimal hyperparameters. BO offers distinct advantages over traditional methods such as grid search, including faster convergence and strong resilience to noisy data [41], which directly addresses the noise-related challenges of this study. The core idea behind BO is to rely on a surrogate function $\mathcal{S}$ to estimate the objective function (*i.e.*, loss function) $\mathcal{F}$, guiding the selection of future sample points. This sampling is regulated by the acquisition function $\mathcal{A}$, which uses the posterior distribution to determine where to explore next. The acquisition function provides a mathematical strategy for navigating the hyperparameter space Θ, based on the predicted mean and variance from the surrogate function, such as the Gaussian Process. In the literature, there are various acquisition functions, but the Expected Improvement (EI) function [42] is the most widely used and has demonstrated high performance in numerous scenarios. Mathematically, the EI function is defined as

$$\text{EI}_{z^*}(h) = \int_{-\infty}^{\infty} \max\left(z^* - z, 0\right) p(z|h) \text{d}z \tag{11}$$

where $p(z|h)$ is the surrogate function, z denotes the true objective function's score, z^* is the maximum score of the observed true objective function, and h represents the hyperparameters.

The pseudocode outlining the use of BO to fine-tune the hyperparameters of the proposed HPE architecture is presented in Algorithm 1.

Algorithm 1: BO-based Hyperparameter Selection

Input: $\mathcal{F}$, $\mathcal{S}_0$, $\mathcal{A}$, $\mathcal{I}$ - the number of iterations
Output: Θ

$\Theta := \varnothing$;
$\iota := 1$;
while $\iota \leq \mathcal{I}$ **do**
  $h^* := \arg\max \mathcal{A}(h, \mathcal{S}_{\iota-1})$;
  Evaluate $f(h^*)$;
  $\Theta := \Theta \cup (h^*, f(h^*))$;
  $\mathcal{S}_\iota$ is trained with Θ;
  $\iota = \iota + 1$;
return Θ

V. Experimental Evaluation

A. Dataset

For our experiments, we exploit the MM-Fi dataset introduced in [43], which is considered the most comprehensive publicly available dataset for wireless HPE. The dataset includes four sensing modalities: RGB-D, LiDAR, mmWave radar, and WiFi. In this study, we focus solely on the WiFi modality to assess the performance of the proposed architecture.

TABLE II: Statistics of the MM-Fi Dataset.

MM-Fi Dataset	Protocol 1	Protocol 2	Protocol 3
Subjects	40	40	40
Transmit antennas	1	1	1
Receive antennas	3	3	3
Subcarriers	114	114	114
Daily activities	14	0	14
Physical therapy exercises	0	13	13
Key-points	17	17	17
Samples	166320	154440	320760

The dimension of CSI data in the MM-Fi dataset is $N_{\text{tx}} \times N_{\text{rx}} \times N_{\text{sc}} \times T \triangleq 1 \times 3 \times 114 \times 10$. As shown in Table II, the data was collected from 40 voluntary subjects (11 females and 29 males). They were assigned to perform 27 action categories: 14 daily movements (corresponding to smart home applications) and 13 physical therapy exercises (corresponding to healthcare applications). Additionally, the MM-Fi dataset provides three different scenarios for evaluation purposes: **Protocol 1** (daily activities), **Protocol 2** (rehabilitation exercises), and **Protocol 3** (a combination of both daily and therapeutic activities). To split the data, one of three methods is utilized: 1) **S1**: Random Split – This method randomly divides all samples into training and testing sets with a $3:1$ ratio, 2) **S2**: Cross-Subject Split – The data is split by subjects, with 32 subjects used for training and 8 for testing, and 3) **S3**: Cross-Environment Split – The data is divided based on the environment, with 3 environments used for training and 1 for testing. These different splitting strategies allow for a comprehensive evaluation of the model. For most of our experiments, we employ the Random Split approach. A performance comparison of these splitting methods is provided in Section V-C.

B. Experimental Setup

1) Data Processing: In practice, CSI data consists of a wide range of amplitude values. Due to the ReLU activation function in the proposed architecture, the Denoiser outputs non-negative

values. Thus, quantizing the data to a uniform range is crucial. In this study, CSI data is normalized to $[0, 1]$ using min-max normalization, which is defined as:

$$v_{\text{norm}} = \frac{v - v_{\text{min}}}{v_{\text{max}} - v_{\text{min}}} \tag{12}$$

where v is a certain value within the range $[v_{\text{min}}, v_{\text{max}}]$.

It is important to note that, while the MM-Fi dataset is heavily pre-processed [43], real-world CSI data often contains noise. Even pre-processed data can still be affected by random noise in practical applications. To simulate more realistic conditions, a percentage of 10% to 30% noise generated from a standard normal distribution is injected into the original dataset. These noisy versions enable a comprehensive validation of the architecture's noise tolerance and adaptability to practical scenarios.

2) Evaluation Metrics: In this study, we use three standard evaluation metrics: Mean Per-Joint Position Error (MPJPE), Procrustes Analysis MPJPE (PA-MPJPE) [44], and Percentage of Correct Key-point (PCK).

The MPJPE metric measures the average Euclidean distance between particular predicted key-points and their corresponding ground truth values. PA-MPJPE is similar to MPJPE but is computed after aligning the points using the Procrustes method. While PA-MPJPE only focuses on the quality of joints, MPJPE evaluates both joint quality and spatial position. It is clear that the lower values of MPJPE and PA-MPJPE signify better HPE performance. Finally, the PCK metric is defined as

$$\text{PCK@}\alpha = \frac{1}{mN} \sum_{i=1}^{N} \sum_{j=1}^{m} \text{LF} \left(\frac{\|pd_{i,j} - gt_{i,j}\|_2^2}{\sqrt{rs^2 + lh^2}} \leq \frac{\alpha}{100} \right) \tag{13}$$

where N is the total amount of data, m is the total number of key-points per label, $\text{LF}(\cdot)$ is a logic function that outputs 1 (true) or 0 (false), and $\alpha(\%)$ is a user-defined threshold to adjust the strictness of the evaluation. The term $\sqrt{rs^2 + lh^2}$ represents the torso size, calculated as the distance between the right shoulder (rs) and the left hip (lh). This measurement is utilized to normalize the prediction error $\|pd_{i,j} - gt_{i,j}\|_2^2$, representing the Euclidean distance between predicted values (pd) and the ground-truth key-points (gt). In the PCK metric, a predicted key-point is considered correct if its distance from the ground truth is within α times the torso size. The recall at a given PCK threshold reflects the percentage of key-points that are correctly identified within that threshold.

3) Implementation Details: The experiments are conducted on an Ubuntu 22.04 workstation equipped with an Intel Core i9-13900K CPU and 32 GB RAM. The Denoiser and Estimator are trained using RMSProp and Adam optimizers, respectively. To ensure consistency across experiments, all models are trained for a maximum of 100 epochs. The batch sizes are set to 256 for the Denoiser and 64 for the Estimator.

C. Results and Discussions

1) Hyperparameter Optimization: This subsection presents the optimal hyperparameters for the MM-Fi dataset determined by the BO algorithm. Table III summarizes the hyperparameters for both the Denoiser and Estimator in the proposed architecture.

TABLE III: Hyperparameters of the Denoiser and Estimator.

	Hyperparameter	Description
Denoiser	*ae_lr*	Learning rate
	ae_momentum	Momentum value
	ae_n_kernels_i	Number of convolutional kernels ($i \in \{1, 2\}$)
	ae_kernel_size_i	Size of respective kernels ($i \in \{1, 2\}$)
	stride_i	Stride of the two convolutions ($i \in \{1, 2\}$)
	ae_maxpooling_i	Size of two max-pooling layers ($i \in \{1, 2\}$)
Estimator	*n_stacks*	Number of stacked dynamic CNN layers
	est_lr	Learning rate
	est_dropout	Drop rate
	est_temp	Temperature of kernel attention to control weight sparsity
	est_n_basis_kernels	Number of basis kernels
	est_kernel_size_i	Size of the basis kernels in the i-th dynamic CNN layer
	est_maxpooling_i	Size of the max-pooling after the i-th dynamic CNN layer
	est_d_hidden	Size of the hidden fully-connected layer

TABLE IV: The Optimal Hyperparameters of the Denoiser.

Hyperparameters	Protocol 1 (10% noise)	Protocol 2 (20% noise)	Protocol 3 (30% noise)
ae_lr	0.0009	0.0007	0.0002
ae_momentum	0.6	0.5	0.5
ae_n_kernels_1	36	58	22
ae_kernel_size_1	3	3	3
ae_maxpooling_1	2	2	2
ae_n_kernels_2	89	144	151
ae_kernel_size_2	3	3	3
ae_maxpooling_2	2	2	2

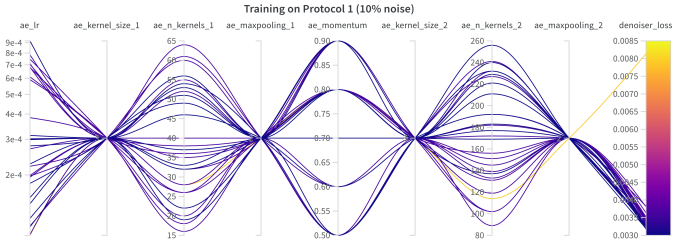
In implementing the BO algorithm, we use log-uniform distributions for continuous hyperparameters and uniform distributions for discrete ones. The optimal hyperparameters for the Denoiser on noisy versions of the MM-Fi dataset are listed in Table IV. Notably, the learning rate of the Denoiser decreases as the noise ratio increases, from 0.0009 at 10% noise to 0.0002 at 30% noise. This is attributed to the fact that a lower learning rate helps stabilize training and improve convergence under higher noise levels, especially in WiFi sensing-based HPE tasks. Additionally, the number of kernels in the second convolutional layer increases significantly, enabling the Denoiser to capture more abstract patterns that are intricately mixed with noise.

Fig. 7 illustrates the BO algorithm's process of automatically searching for the Denoiser's optimal hyperparameters. In this example, the Denoiser is trained on the MM-Fi dataset using Protocol 1 with 10% random noise. Each column in the figure represents a hyperparameter with its respective value ranges. The algorithm clearly adjusts the hyperparameters to minimize the Denoiser's loss. For instance, as shown in Fig. 7(b), the correlation coefficient between the hyperparameter *ae_n_kernels_2* and the Denoiser's loss is -0.224, indicating a negative correlation. Consequently, training with higher *ae_n_kernels_2* values tends to reduce the loss, as shown in Fig. 7(a).

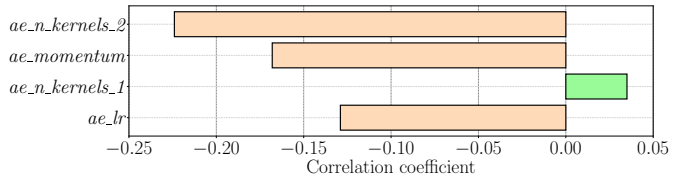
Table V presents the optimal hyperparameters of the Estimator under two scenarios: training on the original MM-Fi

TABLE V: The Optimal Hyperparameters of the Estimator.

Hyperparameter	Protocol 1		Protocol 2		Protocol 3	
	Original	10% noise	Original	20% noise	Original	30% noise
est_lr	0.0049	0.0075	0.0078	0.0010	0.0033	0.0059
est_temp	33	31	32	32	33	30
est_dropout	0.1915	0.2311	0.1361	0.1743	0.1109	0.1346
n_stacks	1	1	1	2	1	3
est_n_basis_kernels	16	205	54	167	44	223
est_n_kernels_1	31	25	24	25	12	16
est_kernel_size_1	3	3	3	2	3	3
est_maxpooling_1	2	2	2	2	3	2
est_n_kernels_2	–	–	–	48	–	19
est_kernel_size_2	–	–	–	2	–	2
est_maxpooling_2	–	–	–	2	–	2
est_n_kernels_3	–	–	–	–	–	101
est_kernel_size_3	–	–	–	–	–	3
est_maxpooling_3	–	–	–	–	–	2
est_d_hidden	256	256	256	32	128	64



(a) Parallel coordinates plot of BO.



(b) Non-zero correlation coefficients with denoiser's loss.

Fig. 7: Hyperparametric search for the Denoiser on Protocol 1 (10% noise).

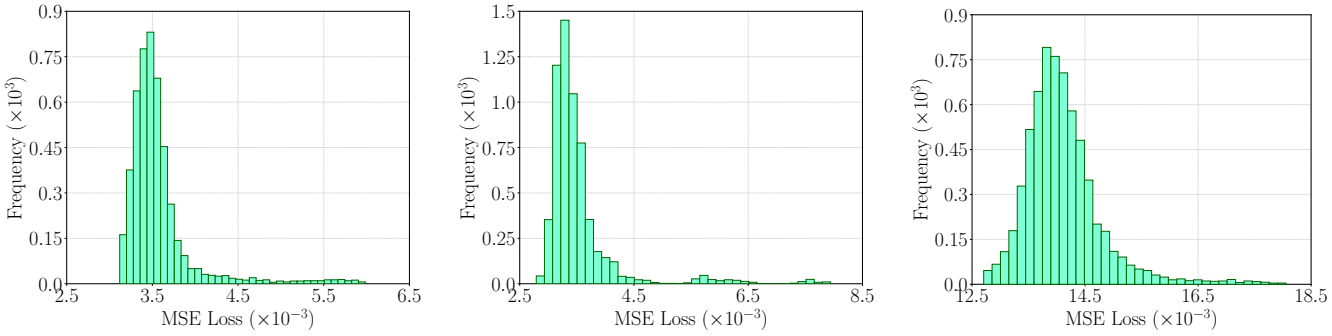
dataset and on its noisy versions. The first scenario allows for a fair comparison of the Estimator's performance with existing methods that do not use a denoising module. The second scenario evaluates and compares the performance of the proposed architecture with previous works that incorporate denoising. As can be seen from the table, the Estimator's depth dynamically adjusts based on the data characteristics. For example, under Protocol 3, the Estimator uses three stacked SDy-CNN networks (*i.e.*, $n_stacks = 3$) when trained on data with 30% noise, compared to just one stack for the original data. This increased depth helps the Estimator better focus on complex CSI features rather than random noise in the data. Additionally, the Estimator employs significantly more basis kernels when trained on noisy data. Similar to the Denoiser, the number of kernels in the initial layers increases with depth, enhancing its ability to capture intricate features. This, thanks to the BO algorithm, highlights the flexibility of the proposed method in adapting the model configurations.

2) Reconstruction of the Denoiser: We inject varying levels of Gaussian noise into the MM-Fi dataset across three different scenarios. The Denoiser takes the noisy data as input and uses clean samples as labels. Its performance is measured by the mean square reconstruction error (MSRE). As shown in Fig. 8, the Denoiser achieves an MSRE close to zero, demonstrating its strong ability to reconstruct the original data.

3) HPE Performance: We now evaluate the HPE performance using the three above-mentioned metrics: PCK, MPJPE, and PA-MPJPE.

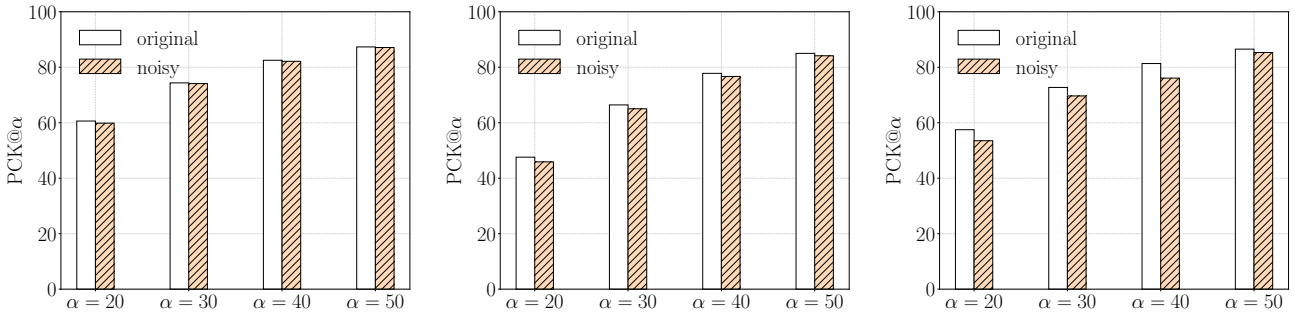
Fig. 9 shows the PCK scores achieved by the proposed architecture across various scenarios involving the original data and its noisy counterparts. The small gaps between PCK@20, PCK@30, PCK@40, and PCK@50 values within each protocol highlight the architecture's robustness, demonstrating its tolerance to random noise and consistency in key-point prediction accuracy under diverse conditions. Notably, the architecture excels in estimating key-points for both daily activities and therapeutic exercises, with PCK@50 exceeding 84% across all scenarios. Even as the number of activities and noise levels increases, the results from Protocol 3 show only slight declines. For example, when the noise ratio rises from 10% in Protocol 1 (14 activities) to 30% in Protocol 3 (27 activities), PCK@50 and PCK@20 decrease only marginally, by about 1.5% (from 87.11% to 85.32%) and 6% (from 59.85% to 53.51%), respectively. It is also worth noting that the accuracy for daily activities slightly outperforms that of therapeutic exercises, with a PCK@50 of 87.11% compared to 84.13%, as shown in Figs. 9(a) and 9(b). This difference is due to the complexity and subtlety of therapeutic movements, such as right upper limb extension or jumping, compared to simpler daily activities like raising an arm or picking up objects.

The quality of joint positions and spatial accuracy of key-points estimated by the proposed architecture is evaluated using MPJPE and PA-MPJPE, with results presented in Table VI. The differences between results on original and noisy data are minimal, even with increased noise levels. For instance, in Protocol 3, which includes both daily activities and therapeutic movements, the architecture achieves an MPJPE of 146.57 mm on the original data and 152.22 mm on the noisy data. The PA-MPJPE values are 91.88 mm and 92.43 mm, respectively. These small gaps align with the PCK scores, further demonstrating the Denoiser's effectiveness in handling random noise



(a) Protocol 1 (10% noise) (b) Protocol 2 (20% noise) (c) Protocol 3 (30% noise)

Fig. 8: Reconstruction accuracy of the Denoiser.



(a) Protocol 1: Original vs. 10% noise (b) Protocol 2: Original vs. 20% noise (c) Protocol 3: Original vs. 30% noise

Fig. 9: The proposed architecture's performance in terms of PCK.

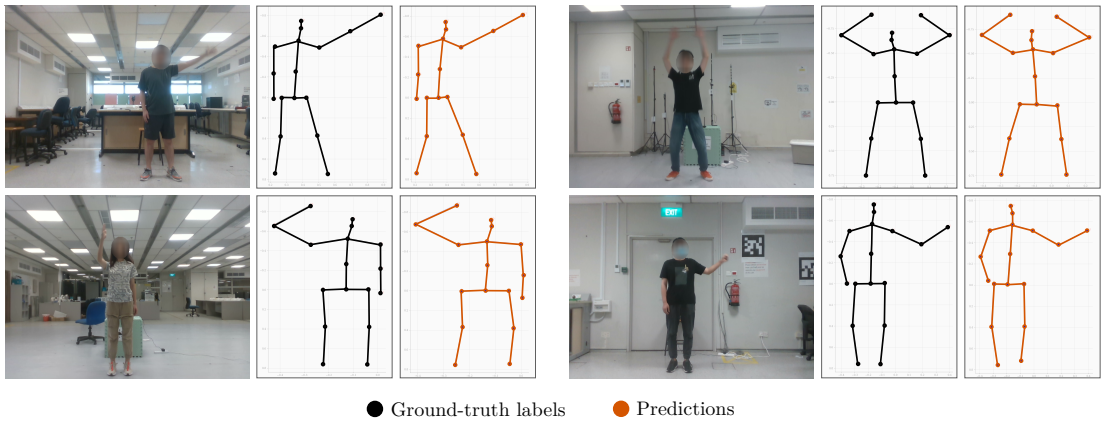


Fig. 10: The key-point visualizations of various subjects in the different environments.

and its contribution to robust HPE performance. For visual presentation, several key-point estimation results are shown in Fig. 10. It can be seen that certain key points are more prone to mislocalization: joints (elbows, knees) due to overlapping body parts, the head due to weak reflections and shoulder interference, and torso midpoints due to overlapping signals from arms or legs.

4) Comparison with Existing Methods: We now compare the performance of the proposed architecture against SOTA models in WiFi sensing-based HPE, including WiSPPN (2019) [12], Mi-Mose (2021) [15], PerUnet (2022) [22], CSIFormer (2022) [18], WPFormer (2023) [11], and CKDformer (2024) [6]. Although most of these models, except for PerUnet and CSIFormer, were designed without considering random noise, we still include them in our comparative analysis. For fairness, all models were reproduced based on their original papers and evaluated on the three protocols of the MM-Fi dataset using the same data-splitting method (S1-Random Split). WiSPPN, Mi-Mose, WPFormer, and CKDformer are compared to our Estimator using the original (clean) data, whereas PerUnet and CSIFormer are compared to our architecture under noisy scenarios. The numerical results are shown in Table VII, with the best performance highlighted in bold and the second-best underlined.

TABLE VI: The Proposed Architecture's Performance in terms of MPJPE and PA-MPJPE.

Metric	Protocol 1		Protocol 2		Protocol 3	
	Original	10% noise	Original	20% noise	Original	30% noise
MPJPE (mm)	133.36	133.56	150.75	155.47	146.47	152.22
PA-MPJPE (mm)	93.51	93.83	85.55	86.84	91.88	92.43

TABLE VII: Performance comparison with SOTA approaches (original and noisy versions). The best performance in original data scenarios and its noisy data scenarios are bolded and underlined, respectively.

Protocol 1 Model	MPJPE (mm) ↓	PA-MPJPE (mm) ↓	PCK@20	PCK@30	PCK@40	PCK@50
WiSPPN	153.72	100.02	49.16	67.07	77.37	83.79
Mi-Mose	152.61	97.28	49.55	67.49	77.72	84.09
WPFormer	140.70	99.20	50.53	69.04	79.48	85.58
CKDformer	145.01	96.28	53.45	69.82	79.19	85.22
Ours	**133.36**	**93.51**	**60.61**	**74.36**	**82.52**	**87.33**
***10%-noise scenario**						
PerUnet	152.16	95.36	50.45	67.79	77.75	84.01
CSIFormer	207.32	101.02	35.95	44.87	59.69	71.63
Ours	<u>133.56</u>	<u>93.83</u>	<u>59.85</u>	<u>74.11</u>	<u>82.14</u>	<u>87.11</u>
Protocol 2 Model	MPJPE (mm) ↓	PA-MPJPE (mm) ↓	PCK@20	PCK@30	PCK@40	PCK@50
WiSPPN	167.88	92.21	39.14	60.11	73.61	82.44
Mi-Mose	170.15	91.81	38.99	59.66	73.00	81.63
WPFormer	191.64	123.64	32.58	51.55	65.75	76.37
CKDformer	169.14	92.13	39.20	60.07	73.17	81.94
Ours	**150.75**	**85.55**	**47.58**	**66.42**	**77.80**	**85.00**
***20%-noise scenario**						
PerUnet	170.67	88.11	39.67	59.66	72.69	81.35
CSIFormer	210.11	96.51	37.10	46.97	62.09	72.08
Ours	<u>155.47</u>	<u>86.84</u>	<u>45.89</u>	<u>65.02</u>	<u>76.69</u>	<u>84.13</u>
Protocol 3 Model	MPJPE (mm) ↓	PA-MPJPE (mm) ↓	PCK@20	PCK@30	PCK@40	PCK@50
WiSPPN	163.19	99.50	45.41	63.21	74.08	80.97
Mi-Mose	155.76	95.35	48.67	66.58	77.34	83.87
WPFormer	152.44	97.44	49.43	67.05	77.45	84.27
CKDformer	147.53	94.20	52.19	68.43	78.13	84.48
Ours	**146.47**	**91.88**	**57.48**	**72.72**	**81.34**	**86.53**
***30%-noise scenario**						
PerUnet	173.53	95.97	39.20	59.08	72.15	80.60
CSIFormer	205.24	98.28	36.96	47.54	62.33	72.46
Ours	<u>152.22</u>	<u>92.43</u>	<u>53.51</u>	<u>69.65</u>	<u>79.09</u>	<u>85.32</u>

Table VII shows that the proposed method significantly outperforms existing benchmarks across all metrics. On the original data, our Estimator achieves remarkable improvements over the best previous model, CKDformer, with PCK@20 and PCK@50 scores increasing by 8.38% and 3.06%, respectively. It also achieves notable reductions in MPJPE and PA-MPJPE errors, outperforming CKDformer by 18.39 mm and 6.58 mm, respectively. WPFormer, in particular, struggles with handling complex poses in Protocol 2 (rehabilitation exercises). This is due to its architecture, which combines a single convolutional network with several Transformer blocks. While CNNs are effective at extracting spatial CSI features, the sequential patterns learned by multiple Transformer blocks with the self-attention mechanism may be less effective for short and complex movements typical in rehabilitation exercises. This observation is supported by results from WiSPPN, Mi-Mose, and CKDformer, which employ more convolutional layers and achieve better outcomes. In contrast, our Estimator, which utilizes SDy-CNN networks, provides the best performance, demonstrating its superior ability to capture significant spatial CSI features.

In noisy scenarios, Table VII demonstrates that the proposed architecture significantly outperforms PerUnet and CSIFormer in terms of denoising. Both PerUnet and CSIFormer rely on multi-head self-attention to focus on undistorted signals for denoising. However, such ideal signals may be rare in real-world environments, leading to diminished performance as noise levels increase. In contrast, the proposed architecture, with its robust denoising capabilities, surpasses the best denoising model, PerUnet, by up to 14%. These results demonstrate the Denoiser's ability to effectively handle random noise, which is a key contribution of the paper.

5) Extended Experiment: As presented in Section V-A, the MM-Fi dataset offers three data-splitting strategies: S1 (Random Split), S2 (Cross-Subject Split), and S3 (Cross-Environment Split). While previous experiments were conducted using the Random Split setting, we now examine the impacts of each data-splitting strategy on the results.

In Table VIII, we provide the MPJPE and PA-MPJPE results obtained by the proposed architecture. A substantial differ-

TABLE VIII: The HPE Performance on the MM-Fi Dataset Using Various Data-splitting Strategies.

Work	Setting	Protocol 1 MPJPE	Protocol 1 PA-MPJPE	Protocol 2 MPJPE	Protocol 2 PA-MPJPE	Protocol 3 MPJPE	Protocol 3 PA-MPJPE
Ours	S1	133.36	93.51	150.75	85.55	146.47	91.88
	S2	169.53	93.89	174.57	86.52	179.18	87.10
	S3	316.94	95.05	316.42	92.34	310.42	91.29
Ours in noisy scenarios*	S1	133.56	93.83	155.47	86.84	152.22	92.43
	S2	170.29	94.20	175.97	87.37	179.47	87.46
	S3	325.95	96.58	319.39	93.84	312.13	94.16

**The noise percentage in noisy scenarios of Protocols 1, 2, 3 are resp.* 10%, 20%, *and* 30%

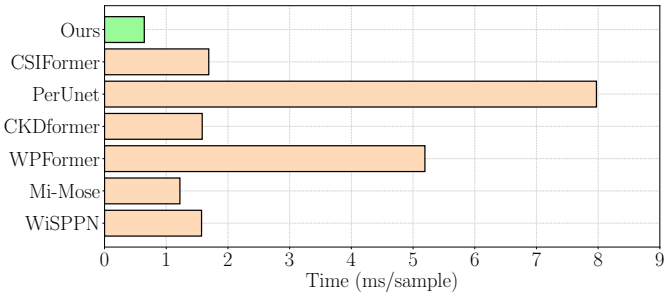


Fig. 11: Average execution time of the different approaches on the MM-Fi dataset.

ence is observed between the Cross-Subject (S2) and Cross-Environment (S3) settings. Specifically, the Cross-Subject results are significantly better than the Cross-Environment results in terms of MPJPE. For example, under noisy conditions, the MPJPE scores for S2 in Protocols 1, 2, and 3 are 120.29 mm, 175.97 mm, and 179.47 mm, respectively, compared to much higher scores of 325.95 mm, 319.39 mm, and 312.13 mm for S3. These significant differences within the same protocol suggest that the proposed architecture is robust to subject variations within a consistent environment (S2). Conversely, changes in the environment (S3) have a more pronounced negative impact on the HPE performance. This challenge is not unique to our approach but it is also observed in other HPE methods, including vision- and radar-based approaches, as noted in [43]. Addressing this impact of environmental changes will be a focus of future research.

6) Empirical Efficiency: Finally, Fig. 11 shows the empirical execution times for various models on the MM-Fi dataset to evaluate their time efficiency. Given that training time costs are typically negligible due to offline computation [17], we focus on the time costs during the testing phase. The results clearly show that self-attention-based methods, such as PerUnet, CSIFormer, and WPFormer are significantly more time-consuming than CNN-based models, including our approaches (*i.e.*, Mi-Mose and WiSPPN). This is due to the added complexity introduced by stacking multiple multi-head self-attention modules alongside CNNs. Among models incorporating denoising, the proposed architecture stands out for its efficiency, being 12.3 times faster than PerUnet and 2.62 times faster than CSIFormer. The time efficiency validates the advantages of the Denoiser design and the proposed SDy-CNN's ability to prioritize relevant subcarriers, enabling better feature extraction while reducing computational overhead. This makes our model particularly well-suited for real-time applications in WiFi sensing-based human pose estimation.

VI. Conclusion

We have proposed a novel neural network architecture combining an AE and SDy-CNN for WiFi sensing-based HPE tasks, aimed at achieving two key objectives: removing unpredictable noise and enhancing pose estimation performance through input-aware adaptability. Using Bayesian optimization, we have leveraged a dynamic neural network and optimized the architecture's configuration. Comparative evaluations highlight the superiority of the proposed method in both denoising and HPE compared to existing models. Specifically, the architecture demonstrates up to a 14% improvement in accuracy and is 12.3 times faster than the previous best model, PerUnet, in denoising scenarios.

The proposed architecture exhibits strong performance and resilience to random noise but presents several areas for improvement. Key challenges include the limited resolution of CSI data, which hinders dense pose estimation, and the lack of reliable public WiFi sensing datasets essential for advancing multi-person pose estimation. Moreover, the Denoiser's reliance on pre-cleaned data highlights the necessity for autonomous denoising techniques capable of managing diverse noise types, while balancing computational efficiency with real-time applicability. While the architecture effectively adapts to short-term changes, addressing long-term environmental and behavioral variations remains challenging. Emerging techniques such as federated learning or continual learning could mitigate these limitations by reducing the reliance on periodic retraining. A novel architecture for multi-user scenarios may benefit from incorporating signal decomposition and spatial filtering techniques to manage overlapping CSI signals, facilitating accurate pose estimation for multiple users. Additionally, enhancing preprocessing pipelines and refining temporal feature weighting is crucial for stability in dynamic environments with non-human interference or moving objects. Future work will prioritize these improvements, scalable designs, and testing on diverse datasets to validate performance in real-world scenarios.

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